

ISO26262 Functional Safety Report

Generated At: 2026-06-20 18:50:34

Development Method: **ATDD**

Field	Value
Document Template	safety_report
Development Method	ATDD
Requirement	what is verification in ISO 26262?
Related ISO Parts	Part 11 - Semiconductors Part 5 - Hardware Development
Referenced Pages	107, 108, 123, 20, 64, 79

ISO26262 Evidence

Part: Part 11 - Semiconductors

Clause: 7.4.2.4

Page: 123

Score: 0.7624

Summary: ■ ISO 26262-11:2018(E) ISO 26262■5:2018 re- quirement Design phase Technique/Measure Aim 7.4.2.4 Robust design principles Synchronisation of primary in- puts and control of metastability Avoidance of ambiguous circuit behaviour as a result of set-up and hold timing violation 7.4.4 Verification of hard- ware design HDL simulation Pre-silicon verification of circuit described in VHDL or Verilog by means of simulation 7.4.4 Verification of hard- ware design Functional test on module level (using for example HDL test benches) Pre-silicon verification "Bot- tom-up" 7.4.4 Verification of hard- ware design Functional test on top level Verification of the PLD (entire function) 7.4.4 Verification of hard- ware design Functional and structural cov- erage-driven verification (with coverage of verific

Part: Part 11 - Semiconductors

Clause: 7.4.4

Page: 107

Score: 0.7443

Summary: ■ ISO 26262-11:2018(E) ISO 26262■5:2018 Clause Design phase Technique/ Measure Aim 7.4.4 Verification of hardware design Requirement Driven Ver- ification All functional and safety-related require- ments are verified. To be shown via traceabili- ty between specification and verification plan 7.4.4 Verification of hardware design Design for testability Specific hardware structures (e.g. test modes, multiplexers) are included into the design and layout in order to test otherwise inaccessible circuit nodes and improve the test coverage 7.4.2.4 Robust design principles Application of schematic design guidelines Manual checks 7.4.4 Verification of hardware design Application of schematic checkers To perform automatic checks for example on interconnections or on the selection of the proper devic

Part: Part 11 - Semiconductors

Clause: 7.4.4

Page: 108

Score: 0.7193

Summary: ■ ISO 26262-11:2018(E) ISO 26262■5:2018 Clause Design phase Technique/ Measure Aim 7.4.4 Verification of hardware design Hardware de- sign verification Development by hard- ware prototyping Verification of implemented functions by prototype (e.g. test chips, boards), can check particular points of the hardware design where design review is not sufficient. 6.5.3 hardware safety require- ment verification report Verification hardware safety require- ment verification report Provide evidence of consistency with hardware specification, completeness and correctness 10.5.1 hardware integration and verification activ- ities Hardware integration ver- ification Verification of the completeness and cor- rectness of the design implementation on the component level Perform component tests and reports

Part: Part 11 - Semiconductors

Clause: 7.4.4

Page: 79

Score: 0.7034

Summary: ■ ISO 26262-11:2018(E) ISO 26262-5:2018 requirement Design phase Technique/Measure Aim 7.4.4 Verification of hardware design Pre-silicon verification on top level Verification of the entire circuit. 7.4.2.4 Robust design principles Restricted use of asynchronous constructs Avoidance of typical timing anomalies during synthesis, avoidance of ambiguity during simulation and synthesis caused by insufficient modelling, design for testability. This does not exclude that for certain types of circuitry, such as reset logic or for very low-power microcontrollers, asynchronous logic could be useful: in this case, the aim is to suggest additional care to handle and verify those circuits. 7.4.2.4 Robust design principles Synchronisation of primary inputs and control of meta-stability Avoidance

Part: Part 5 - Hardware Development

Clause: 4.5.4.3

Page: 20

Score: 0.6953

Summary: ■ ISO 26262-11:2018(E) plans are included for applicable supporting processes as described in ISO 26262-8, covering configuration management, change management, impact analysis and change requests, verification, documentation management and software tool qualification. 4.5.4.3 Safety requirements allocated to the IP design The hardware safety requirements can be allocated to the IP design as defined in ISO 26262-5:2018, Clause 6. EXAMPLE The requirement for a safety mechanism in the IP is described, allowing the requirement to be verified at an appropriate level of integration. The integration and test specifications can be linked to requirements defined in the technical safety concept. 4.5.4.4 Hardware design verification and verification review of the IP design Defining criteria for desi

Part: Part 11 - Semiconductors

Clause: Unknown

Page: 64

Score: 0.6899

Summary: ■ ISO 26262-11:2018(E) ■ type of safety mechanism including required confidence level; ■ observation points and diagnostic points; ■ fault site, fault list; and ■ workload used during fault injection. In particular, the verification planning describes and justifies: ■ the fault model and the related level of abstraction: ■ As clarified in the following clauses for DFA, digital, analogue and PLD, fault injection can be performed at the appropriate level depending on the fault model being considered, the specific semiconductor technology, feasibility, observability and use case; and NOTE 1 Depending on the purpose, fault injection can be implemented at different abstraction levels (e.g. semiconductor component top-level, part or subpart level, RTL, etc.). A rationale for the abstraction leve

Hazard / ASIL

Timing violations leading to metastability in hardware design

ASIL: N/A

Safety Goal / FSR / TSR

Safety Goal:

Ensure hardware design meets safety requirements as specified in technical safety concept

FSR:

Hardware safety requirement verification report

TSR:

Functional and structural coverage-driven verification

ATDD Acceptance Criteria

TC-001 - Verify hardware design functionality through HDL simulation

Given: Hardware design is implemented in HDL (VHDL/Verilog)

When: Pre-silicon simulation is run on the top-level circuit
Then: Functional tests pass with at least 90% coverage of safety requirements
Expected Action: Generate verification report with coverage metrics
Verification Method: HDL simulation, functional test, coverage analysis
Methodology: ATDD

Traceability

Development Method: ATDD
Requirement: REQ-001
Hazard: HZ-001
Safety Goal: SG-001
FSR: FSR-001
TSR: TSR-001
Test Cases: TC-001

Trace Chain:
REQ-001 → HZ-001 → SG-001 → FSR-001 → TSR-001 → TC-001